

INTELLECTUAL PROPERTY RIGHTS AND PATENTS

UNIT – I Definitions and importance of intellectual property – Introduction and history of WTO – Structure of WTO agreements and dispute settlements – Principles of trading system – Trade policy reviews – Agreement on TRIPS – Ministerial conferences – Emerging issues in IPR – Protection of plant varieties – Patent sharks – Open-source movement – Bio-piracy

1.1 Definitions and importance of intellectual property :

Definition of Intellectual Property :

According to WIPO

Intellectual property (IP) refers to creations of the mind, such as inventions; literary and artistic works; designs; and symbols, names and images used in commerce.

Intellectual property (IP) contributes enormously to our national and state economies. Dozens of industries across our economy rely on the adequate enforcement of their patents, trademarks, and copyrights, while consumers use IP to ensure they are purchasing safe, guaranteed products. We believe IP rights are worth protecting, both domestically and abroad. This is why:

Importance of Intellectual Property

- **Creates and Supports High-Paying Jobs**

IP-intensive industries employ hundreds of millions of people worldwide.

Jobs in IP-intensive industries are expected to grow faster over the next decade than the national average.

The average worker in an IP-intensive industry earned about 46% more than his counterpart in a non-IP industry. Eg; TCS in India

- **Drives Economic Growth and Competitiveness**

The nation's economy and GDP is determined through Intellectual Property. America GDP is highest in the world because of its Intellectual Property. The Intellectual property leads to economic growth and employment.

- **Protect Consumers and Families**

Strong IP rights help consumers make an educated choice about the safety, reliability, and effectiveness of their purchases.

Enforced IP rights ensure products are authentic, and of the high-quality that consumers recognize and expect.

IP rights foster the confidence and ease of mind that consumers demand and markets rely on.

- **Generate Solutions to Global Challenges**

Nearly all of the hundreds of products on the World Health Organization's Essential Drug List, which are critical to saving or improving people's lives around the globe, came from the R&D-intensive pharmaceutical industry that depends on patent protections.

Innovative agricultural companies are creating new products to help farmers produce more and better products for the world's hungry while reducing the environmental impact of agriculture.

IP-driven discoveries in alternative energy and green technologies will help improve energy security and address climate change.

- **Encourage Innovation and Reward Entrepreneurs**

Risk and occasional failure are the lifeblood of the innovation economy. IP rights incentivize entrepreneurs to keep pushing for new advances in the face of negative outcomes.

IP rights facilitate the free flow of information by sharing the protected know-how critical to the original, patented invention. In turn, this process leads to new innovations and improvements on existing ones.

Americans's Founding Fathers so recognized the importance of innovation and ensured that strong IP rights for authors and inventors are protected in the U.S. Constitution, thus making America the world's entrepreneurial leader—a fact borne out by the overwhelming number of patents, copyrights and trademarks filed by the U.S. annually.

1.2 Introduction and history of WTO

The World Trade Organization (WTO) is the global international organization dealing with the rules of trade between nations. WTO agreements are negotiated and signed by countries which are members of WTO. The goal is to help producers of goods and services, exporters, and importers conduct their business.

Evolution and History of WTO :

1. Trade Agreement:

In 1934, the United States passed the Reciprocal Trade Agreements Act of 1934 for promoting and regulating international commerce. Bilateral trade agreements. (Bilateral agreements are those between two parties — for example, the US and another country.) were negotiated in the hopes of expanding employment. During the 1930s, the amount of bilateral negotiation under this act was fairly limited,

2. Establishment of ITO :

In the 1940s, working with the British government, the United States developed ITO (International Trade Organization). The **International Trade Organization (ITO)** was the proposed name for an international institution for the regulation of trade.

Led by the United States in collaboration with allies, the effort to form the organization from 1945 to 1948, with the successful passing of the Havana Charter, eventually failed due to lack of approval by the US Congress.

3. GATT

The General Agreement on Tariffs and Trade (GATT), signed in 1947 by 23 countries, is a treaty minimizing barriers to international trade by eliminating or reducing quotas, tariffs, and subsidies. It was intended to boost economic recovery after World War II

4. The Establishment of the WTO after Uruguay Round:

By the late 1980s, a growing number of nations decided that GATT could better serve global trade expansion if it became a formal international organization. An effective dispute settlement mechanism was the need of the hour. They pressed for negotiations to formalize GATT and to make it a more powerful and comprehensive organization. The result was the World Trade Organization, (WTO), which was established during the Uruguay Round (1986-1993) of GATT negotiations and which subsumed GATT. The WTO began life on 1 January 1995. The WTO provides a permanent arena for member governments to address international trade issues and it oversees the implementation of the trade agreements negotiated in the Uruguay Round of trade talks.

Objectives and functions of World Trade Organisation (WTO)

The objectives of World Trade Organisation (WTO) are:

1. Creating and Enforcing International Trade Regulations:

The General Agreement on Trade in Services, the Trade-Related Aspects of Intellectual Property Rights Agreement, and the Agreement on International Trade in Goods, all serve as the foundation for the World Trade Organization (WTO). The WTO uses a multilateral dispute settlement system to enforce its rules when one of its member countries violates a trade agreement. The methods and decisions must be respected and adhered to by the members through signed agreements.

2. Making the Decision Making Process More Transparent:

The WTO has made an effort to promote transparency in decision-making by encouraging participation and, in particular, the use of the consensus rule. Such measures work together to increase institutional transparency.

3. Collaboration between International Economic Institutions:

The onset of globalization has made strong collaboration amongst multilateral institutions necessary. The World Trade Organization, the International Monetary Fund, the United Nations Conference on Trade and Development, and the World Bank are some international economic institutions. These institutions help develop and carry out a framework for international economic policy. Policy making may be disturbed in the absence of regular cooperation and mutual participation.

4. Serving as the World's Leading Forum:

The WTO is the international platform for regulating and negotiating additional trade liberalization. The foundation of WTO liberalization initiatives is based on members' benefits to make the best use of their comparative advantages as a result of a free and fair trade system.

5. Settlements of Trade Disputes:

Before the WTO, trade disputes usually arise from the breach of agreements between the member nations. Such trade disputes are settled through a multilateral system with predetermined rules and regulations.

6. Others:

Some of the other objectives of the World Trade Organisation are as follows:

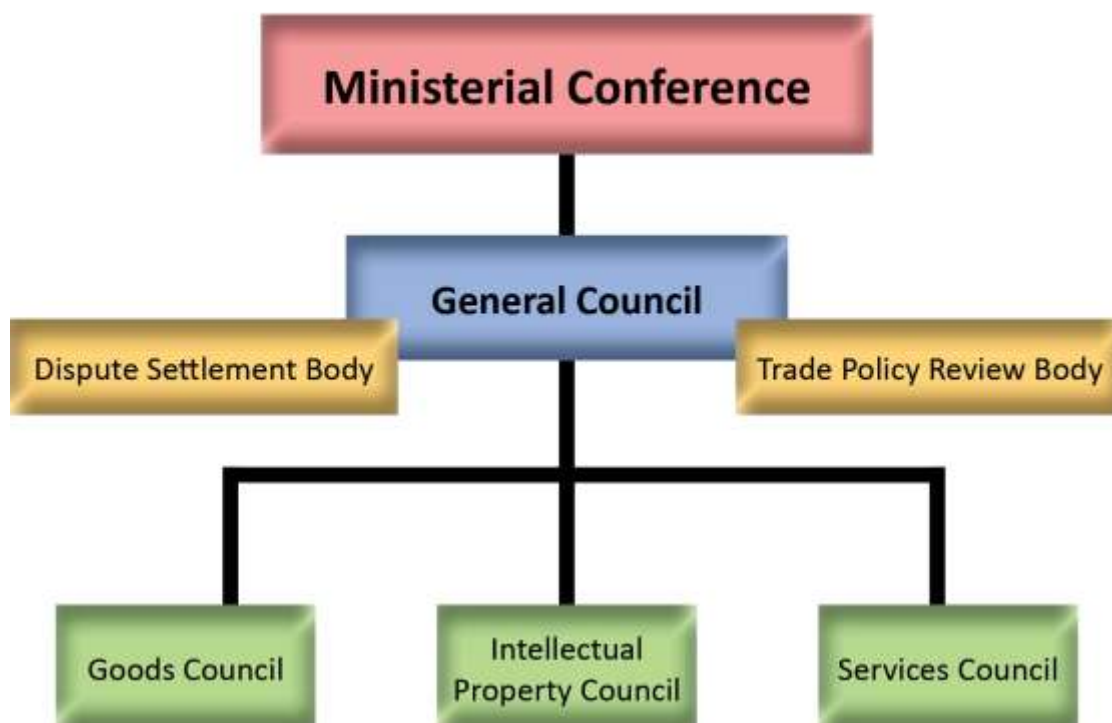
- To ensure optimum utilization of world resources.
- To protect the environment.
- To ensure full employment and a significant rise in effective demand.
- To raise the level of standard of living for citizens of member nations.
- To embrace the idea of sustainable development.

1.3 : Structure of WTO and its Agreements

- Structure of WTO

The WTO is run by its member governments. All major decisions are made by the membership as a whole, either by ministers (who meet at least once every two years) or by their ambassadors or delegates (who meet regularly in Geneva). Decisions are normally taken by consensus.

Structures of WTO



- Highest authority: the Ministerial Conference So, the WTO belongs to its members. The countries make their decisions through various councils and committees, whose membership consists of all WTO members. Topmost is the ministerial conference which has to meet at least once every two years. The Ministerial Conference can take decisions on all matters under any of the multilateral trade agreement
- Second level: General Council Day-to-day work in between the ministerial conferences is handled by three bodies:
 - The General Council
 - The Dispute Settlement Body
 - The Trade Policy Review Body

All three are in fact the same — the Agreement Establishing the WTO states they are all the General Council, although they meet under different terms of reference. Again, all three consist of all WTO members. They report to the Ministerial Conference. The General Council acts on behalf of the Ministerial Conference on all WTO affairs. It meets as the Dispute Settlement Body and the Trade Policy Review Body to oversee procedures for settling disputes between members and to analyze members' trade policies.

- Third level: Three more councils, each handling a different broad area of trade, report to the General Council:
 - The Council for Trade in Goods (Goods Council)
 - The Council for Trade in Services (Services Council)
 - The Council for Trade-Related Aspects of Intellectual Property Rights (TRIPS Council)As their names indicate, the three are responsible for the workings of the WTO agreements dealing with their respective areas of trade. Again they consist of all WTO members. . They cover issues such as trade and development, the environment, regional trading arrangements, and administrative issues.

1.4 The Dispute Settlement Body (DSB)

The composition:

The General Council of WTO discharges its responsibilities under the Dispute Settlement Body. DSB is composed of representatives of all WTO Members.

These are governmental representatives, in most cases diplomatic delegates who reside in Geneva (where the WTO is based) and who belong to either the trade or the foreign affairs ministry of the WTO Member they represent. As civil servants, they receive instructions from their capitals on the positions to take and the statements to make in the DSB. As such, the DSB is a political body.

Functions of DSB:

The DSB is responsible for overseeing the entire dispute settlement process.

Establish panels: .

The DSB has the authority to establish panels, adopt panel and Appellate Body. The panel has to prepare reports and implement the rules and regulations with regard to dispute settlement.

Conduct meeting:

DSB usually has one regular meeting per month. When a Member so requests, the Director-General convenes additional special meetings. The staff of the WTO Secretariat provides administrative support for the DSB

Adjudication and decision by consensus:

In case of disputes, DSB is responsible for the referral of a dispute to adjudication (establishing a panel); for making the adjudicative decision binding (adopting the reports); generally, for supervising the implementation of the ruling; and for authorizing “retaliation” when a Member does not comply with the ruling.

Consensus decision: The general rule is for the DSB to take decisions by consensus. DSU defines consensus as being achieved if no WTO Member, present at the meeting when the decision is taken, formally objects to the proposed decision. This means that the chairperson does not actively ask every delegation whether it supports the proposed decision, nor is there a vote. On the contrary, the chairperson merely asks, for example, whether the decision can be adopted and if no one raises their voice in opposition, it is consensus.

the chairperson will announce that the decision has been taken or adopted. In other words, a delegation wishing to block a decision is obliged to be present and alert at the meeting, and when the moment comes, it must raise its flag and voice opposition. Any Member that does so, even alone, is able to prevent the decision.

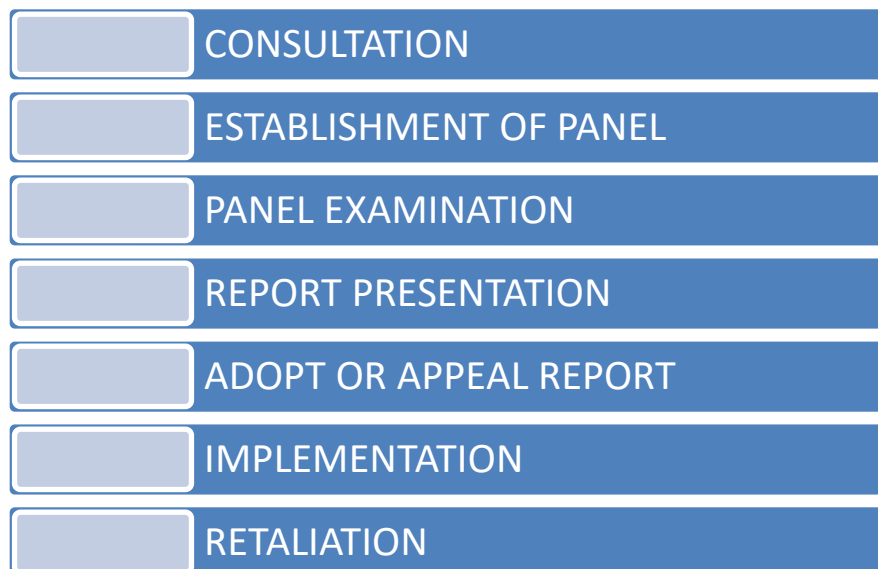
Ensure full participation:

No Member (including the affected or interested parties) is excluded from participation in the decision-making process. This means that the Member requesting the establishment of a panel, the adoption of the report

Stages in a typical WTO dispute settlement case

A dispute can pass in the (WTO) dispute settlement system. There are two main ways to settle a dispute once a complaint has been filed in the WTO: (i) the parties find a mutually agreed solution, particularly during the phase of bilateral consultations; and (ii) through adjudication, including the subsequent implementation of the panel and Appellate Body reports, which are binding upon the parties once adopted by the DSB. There are three main stages to the WTO dispute settlement process:

- (i) consultations between the parties;
- (ii) adjudication by panels and, if applicable, by the Appellate Body; and
- (iii) the implementation of the ruling, which includes the possibility of countermeasures in the event of failure by the losing party to implement the ruling.



1.5 Principles of trading system –

The multilateral trading system of WTO follows certain principles

a. Trade without discrimination

1. Most-favoured-nation (MFN): treating other people equally Under the WTO agreements, countries cannot normally discriminate between their trading partners. Grant someone a special favour (such as a lower customs duty rate for one of their products) and you have to do the same for all other WTO members.

2. National treatment: Treating foreigners and locals equally Imported and locally-produced goods should be treated equally — at least after the foreign goods have entered the market. The same should apply to foreign and domestic services, and to foreign and local trademarks, copyrights and patents. This principle of “national treatment” (giving others the same treatment as one’s own nationals) is also found in all the three main WTO agreements

Freer trade: gradually, through negotiation

Lowering trade barriers is one of the most obvious means of encouraging trade. The barriers concerned include customs duties (or tariffs) and measures such as import bans or quotas that restrict quantities selectively. From time to time other issues such as red tape and exchange rate policies have also been discussed.

Predictability: through binding and transparency

Sometimes, promising not to raise a trade barrier can be as important as lowering one, because the promise gives businesses a clearer view of their future opportunities. With stability and predictability, investment is encouraged, jobs are created and consumers can fully enjoy the benefits of competition — choice and lower prices. The multilateral trading system is an attempt by governments to make the business environment stable and predictable.

Promoting fair competition

The WTO is sometimes described as a “free trade” institution, but that is not entirely accurate. The system does allow tariffs and, in limited circumstances, other forms of protection. More accurately, it is a system of rules dedicated to open, fair and undistorted competition.

The rules on non-discrimination — MFN and national treatment — are designed to secure fair conditions of trade. So too are those on dumping (exporting at below cost to gain market share) and subsidies. The issues are complex, and the rules try to establish what is fair or unfair, and how governments can respond, in particular by charging additional import duties calculated to compensate for damage caused by unfair trade.

Encouraging development and economic reform

The WTO system contributes to development. On the other hand, developing countries need flexibility in the time they take to implement the system’s agreements. And the agreements themselves inherit the earlier provisions of GATT that allow for special assistance and trade concessions for developing countries.

Over three quarters of WTO members are developing countries and countries in transition to market economies. During the seven and a half years of the

The trading system should be ...

- without discrimination — a country should not discriminate between its trading partners (giving them equally “most-favoured-nation” or MFN status); and it should not discriminate between its own and foreign products, services or nationals (giving them “national treatment”);
- freer — barriers coming down through negotiation;
- predictable — foreign companies, investors and governments should be confident that trade barriers (including tariffs and non-tariff barriers) should not be raised arbitrarily; tariff rates and market-opening commitments are “bound” in the WTO;
- more competitive — discouraging “unfair” practices such as export subsidies and dumping products at below cost to gain market share;
- more beneficial for less developed countries — giving them more time to adjust, greater flexibility, and special privileges.

1.6 Trade policy reviews

Trade Policy Review Mechanism

- The **Trade Policy Review Mechanism (TPRM)** was an early result of the **Uruguay Round**.
- It is the **main transparency instrument of the WTO**, affording opportunities for a process of collective evaluation of the trade policies and practices of individual members.
- **Objectives:**
 - Facilitating the smooth functioning of the multilateral trading system by enhancing the transparency of Members’ trade policies.
 - To examine the impact of a Member’s trade policies and practices on the multilateral trading system.
- **Mechanism:**
 - The reviews **take place in the Trade Policy Review Body** which is actually the WTO General Council — comprising the WTO’s full membership — operating under special rules and procedures.
- **Function:**
 - The trade policy review **allows members to put the overall trade and economic policies of a country under the scanner**.
 - The trade policies of developing countries are taken up for review every four years while developed ones face similar scrutiny every two years.
 - The mandate of the TPRM was **broadened to cover services trade and intellectual property**.
- **All WTO Members are subject to review** under the TPRM.

The final session of India’s seventh **Trade Policy Review (TPR)** concluded at the **World Trade Organization (WTO)** in Geneva, Switzerland during 2021.

Key Points

- **Appreciation for India:**

- Introduction of **Goods & Services Tax**
- India's efforts in the implementation of **WTO's Trade Facilitation Agreement**.
- India's improved ranking in "**Trading across Borders**" indicator under the **Ease of Doing Business Report**.

India's **National Intellectual Property Rights Policy, 2016**.

- **Concerns for India:**

- India's trade **policy remained largely unchanged** since the previous review.
- India continues to **rely on trade policy instruments** such as the **tariff, export taxes, minimum import prices, import and export restrictions, and licensing**, WTO said.

1.7 Agreement on TRIPS

Overview: the TRIPS Agreement

The TRIPS Agreement, which came into effect on 1 January 1995, is to date the most comprehensive multilateral agreement on intellectual property.

INTELLECTUAL PROPERTY RIGHTS Introduction Intellectual property (IP) is a term referring to creation of the intellect (the term used in studies of the human mind) for which a monopoly (from greek word monos means single polein to sell) is assigned to designated owners by law. Some common types of intellectual property rights (IPR), in some foreign countries intellectual property rights is referred to as industrial property, copyright, patent and trademarks, trade secrets all these cover music, literature and other artistic works, discoveries and inventions and words, phrases, symbols and designs. Intellectual Property Rights are themselves a form of property called intangible property. .

The three main features of the Agreement are:

- **Standards.** In respect of each of the main areas of intellectual property covered by the TRIPS Agreement, the Agreement sets out the minimum standards of protection to be provided by each Member. Each of the main elements of protection is defined, namely the subject-matter to be protected, the rights to be conferred and permissible exceptions to those rights, and the minimum duration of protection.
- **Enforcement.** The second main set of provisions deals with domestic procedures and remedies for the enforcement of intellectual property rights. The Agreement lays down certain general principles applicable to all IPR enforcement procedures
- **Dispute settlement.** The Agreement makes disputes between WTO Members about the respect of the TRIPS obligations subject to the WTO's dispute settlement procedures.

Scope of TRIPS agreement/ Types of Intellectual Property:

a. Copyrights and related rights

The Agreement states that copyright protection only applies to phrases, ideas, techniques, operating methods, or mathematical concepts. Literary, musical, dramatic, photographic, sculptural, architectural, choreography, graphic, motion picture, sound recording, multimedia work, computer programs, and other works are all given copyright. For a certain amount of time, the owner of a copyright has the right to prevent others from duplicating, distributing, making derivative works, performing, exhibiting, or utilising the work covered by the copyright. The

essence of copyright is originality, which means that the work was created by the copyright owner or claimant. A work of originality, on the other hand, does not have to be innovative. In copyright law, originality does not entail innovation.

b.Trademarks

Trademark is any sign, or set of signs, able to distinguish one undertaking's products and services from other undertakings', shall be eligible for trademark registration, provided that it is clearly detectable. Such signs, in particular words, characters, digits, figurative components, and colour combinations, as well as any combination of these signs, must be acceptable for trademark registration. The trademark owner has the exclusive right to restrict third parties from using similar or identical signs for products or services that are similar to those for which the trademark is registered.

c.Geographical indications

Geographical indications designate a good as coming from a member's territory, or an area or place within that territory, where the good's quality, reputation, or other attribute is largely due to its geographical origin.

d.Industrial designs

The member countries must ensure that fresh or unique industrial designs generated independently are protected. When such activities are conducted for commercial objectives, the right holder can ban third parties who do not have the holder's agreement from producing, importing or selling items that incorporate the protected design.

e. Patents

A patent is an intellectual property right (IPR) awarded to inventors. The inventor, as the patent owner, has the right to prevent anybody else from creating, using, selling, or importing the patent-protected invention in a specified region for a set length of time.

f. Layout-Designs of Integrated Circuits

Importing, selling, or distributing (for commercial reasons) a secured layout design, an integrated circuit where a secured layout design is implemented, or an article including such a circuit is prohibited

g. Protection of undisclosed information

The information which is undisclosed is referred to as a trade secret . It provide trade secret protection in accordance with the Agreement's provisions.

1.8 Ministerial conferences

The Ministerial Conference, the WTO's highest decision-making entity, convenes typically every two years. It puts together all of the WTO's members, which are either nations or customs unions. All decisions pertaining to any of the multilateral trade agreements must be made by the Ministerial Conference.

About WTO Ministerial Conferences

The World Trade Organization's highest competent authority is the Ministerial Conference (WTO). From 1996 until 2022, there were twelve ministerial conferences, which were typically held every two years. Below are some brief details about all such twelve conferences held so far:

First Ministerial Conference

In 1996, Singapore hosted the first ministerial meeting. Its main goal was to launch a worldwide initiative among trading nations to reform the General Agreement on Tariffs and Trade

Second Ministerial Conference

During , 1998, Geneva, Switzerland hosted the 2nd WTO Ministerial Conference. The fierce competition between huge corporations and between countries in the great race of globalisation had been discussed in this conference.

Third Ministerial Conference

In 1999, USA hosted the 3rd WTO Ministerial Conference. The conference's main objective was to make a resolution regarding the start of the so-called "New Rounds,"

Fourth Ministerial Conference

The Fourth Ministerial Conference in Doha, Qatar, declared in November 2001 that negotiations on a variety of topics and other tasks, such as concerns about the execution of the current accords,

.

Fifth Ministerial Conference

From September 10 to 14, 2003, Mexico hosted the 5th WTO Ministerial Conference. The major goal was to evaluate the state of the . It accepted Cambodia and Nepal as the WTO's first newly admitted least-developed nations since the organization's founding.

Sixth Ministerial Conference

From December 13 to December 18, 2005, Hong Kong hosted the 6th WTO Conference Ministerial.. At this summit, nations decided to phase out all export subsidies for agriculture by the end of 2013

Seventh Ministerial Conference

From 30th November to 2nd December 2009, Geneva, Switzerland, hosted the WTO Ministerial Conference's 7th Session. "The WTO, the Multilateral Trading System, and the Current Global Economic Environment" was the overarching theme of discussion.

Eighth Ministerial Conference

From December 15th to December 17th, 2011, Geneva, Switzerland, hosted the 8th ministerial conference. Three working sessions with the topics of "Importance of the Multilateral Trading System and the WTO," "Trade and Development,"

Ninth Ministerial Conference

A set of decisions intended at streamlining trade, giving developing nations more choices for ensuring food security, enhancing trade with least-developed nations, and generally assisting development, was adopted by ministers at the 9th Ministerial Conference, hosted in Bali, Indonesia, from 3rd to 7th December 2013

Tenth Ministerial Conference

From December 15 to 19, 2015, Nairobi, Kenya, hosted the 10th Ministerial Conference of the World Trade Organization. It culminated," a collection of 6 Ministerial Decisions on matters pertaining to least-developed nations, cotton, and agriculture (LDCs)..

Eleventh Ministerial Conference

Argentina, hosted the 11th Ministerial Conference, the Conference concluded with a series of ministerial decisions, including ones on e-commerce duties and fisheries subsidies.

Twelfth Ministerial Conference

From June 12 to June 17, 2022, the WTO's twelfth Ministerial Conference was held in Geneva. Ministers from all over the world were present to examine how the multilateral trading system was operating, make general remarks, and decide how the WTO should proceed with its future activities.

Important Decisions of Twelfth WTO Ministerial Conference

Curtailing Harmful Subsidies in the Fisheries Sector

E-commerce Transactions

COVID-19 Vaccine

1.9 Emerging issues in IPR –

Emerging Issues in Intellectual Property

The emerging issues have been listed below.

- **Artificial intelligence and machine learning**

The point is explained below.

- Artificial intelligence systems are forging patented inventions, copyrighted works, and trademarks at an increasing rate
- There is uncertainty regarding who owns the IP rights for such creations - the AI creator, developer or user
- Most nations have not set clear laws managing AI intellectual property rights
- This has led to legal fuss over who can claim ownership and benefit from AI-generated creations.

- **Digital technologies**

The point is explained below.

- Technologies like 3D printing, online content sharing, and blockchain pose a fuss in setting IP laws
- 3D printing enables easy, large-scale image of patented designs, which is hard to monitor or enforce
- Content-sharing media make it hard to track copyright breaches and enforce takedown requests

- **Knowledge economy**

The point is explained below.

- In the knowledge economy, the boundaries between basic research and commercial products/services are blurring
- This raises concerns that extensive IP defense could restrict access to learning and research that could benefit society
- There are calls for IP law reforms to achieve a better balance between the commercial interests of IP owners and the public interests of access to learning.

- **Globalization**

There is a need to harmonize IP laws and frameworks across nations to promote global trade, investment, and innovation. Yet, many nations approach IP differently based on economic growth priorities and interests. This gap has stalled, telling international IP law harmonization.

- **Public interest**

Intellectual property rights meant to incentivize creation and creation are being monitored if they disproportionately help firms at the cost of public interests.

- Strict IP protections are argued to raise prices, restrict access and create monopolies against the public interest.
- This has led to demands for IP law reforms that achieve a more equitable balance between commercial and non-commercial interests of knowledge sharing and access.

- **Economic growth**

Many developing nations argue that current IP laws favor set nations and deny their ability to utilize their ability for economic development.

- Strict IP protections prevent access to technologies, knowledge, and cultural effects for growth.
- This has stalled international IP law harmonization as forging nations lobby for more flexible IP regimes that support their foci.

Conclusion

While intellectual property rights aim to enable creation, several issues pose new fuss for IP regimes worldwide. Technological advances, financial changes, and the need for sustainable development demand timely and level reforms of IP laws and policies. While several nations are toiling to update IP laws to handle arising issues, useful keys are still lacking. Wide reforms must recognize the tricky interplay amongst IP, creation, access, trade and sustainable growth. The key is flat IP policies that enable the long-term gain of learning and interest for all.

1.10 Protection of plant varieties ;

PLANT VARIETY PROTECTION

Plant variety protection provides legal protection of a plant variety to a breeder in the form of Plant Breeder's Rights (PBRs). PBRs are intellectual property rights that provide exclusive rights to a breeder of the registered variety. In India, the Plant Variety Protection And Farmers Rights (PPVFR) Act, 2001 is a *sui generis* system that aims to provide for the establishment of an effective system for protection of plant varieties and the rights of plant breeders and farmers.

A certificate of registration for a variety issued under this Act confers an exclusive right on the breeder or his successor, his agent or licensee, to produce, sell, market, distribute, import or export the variety. Application for registration of plant varieties can be made in the office of Registrar, PPV & FRA, New Delhi.

Registrable Plant Varieties in India

The following types of plant varieties can be registered under PPVFR Act, 2001:

- New varieties
- Extant variety
- Farmers' variety
- Essentially derived variety

A new variety shall be registered under PPVFR Act, 2001 if it conforms to the criteria of novelty, distinctiveness, uniformity and stability.

Non- Registrable Plant Varieties in India

The following types of plant varieties cannot be registered under the PPVFR Act, 2001:

- Any variety where prevention of commercial exploitation of such variety is necessary to protect public order or public morality or human, animal and plant life and health or to avoid serious prejudice to the environment.
- Any varieties that involves any technology which is injurious to the life or health of human beings, animals or plants. The expression “any technology” includes genetic use restriction technology and terminator technology.
- Variety belonging to the species or genera which is not listed in the notification issued by the Central Government.

Duration of Protection for a Registered Variety

- **Trees and vines** – 18 years from the date of registration of the variety.
- **Extant varieties** – 15 years from the date of notification of that variety by the Central Government under Seed Act, 1966.
- **Other crops** – 15 years fifteen years from the date of registration of the variety.

1.11 Patent sharks –

Patent Sharks

What’s a Patent Shark?

To be clear, patent sharks are not inventors who pursue their own line of research, visibly offer it for sale or licensing early on in the process, and then defend their rights against deliberate infringements.

Instead, they are entities that—opportunistically or intentionally—profit from payments by companies that inadvertently infringe on the sharks’ IP rights simply because they never even knew those rights existed.

Case Example of Patent Shark:

. Chip maker Intel, for example, has had to pay Intergraph at least \$675 million for infringement of its Clipper processor patents—even though Intergraph stopped manufacturing hardware years before and has never used some of the patents it holds in its products.

Case Example of Asure:

Another shark, Asure Software (formerly known as Forgent Networks), has also realized enormous revenues: more than \$100 million from a single (acquired) patent.

Mechanism adopted by Patent Sharks:

a. Preliminary injunctions.

Sharks can put enormous pressure on manufacturers by threatening to shut down their operations using a preliminary injunction. This tactic is especially effective in a fast-paced, complex industry like mobile phone operations, where closing the doors—no matter how briefly—can put even an established player out of business.

Eg: Consider what happened to Research in Motion (RIM), which builds the best-selling BlackBerry and runs the mobile e-mail service of the same name. NTP, a pure patent-holding

company, first filed suit against RIM in 2000 for violation of five of its patents and petitioned the court for an injunction on the sale and support of BlackBerry devices.

b. Damages.

Sharks can also win excessive damages at the end of an infringement trial.

The mechanism adopted by Patent Shark is

Hide – Seek – Sue

Hide : Obtain patent and hide themselves from the business operations for some time

Seek : After some years identify the organization using their patented inventions

Sue : File a case and claim huge damages for infringement of patent.

.

1.12 Open-source movement – Bio-piracy

Open source software movement

The **open-source-software movement** is a movement that supports the use of open-source licenses for some or all software, as part of the broader notion of open collaboration. The open-source movement was started to spread the concept/idea of open-source software.

Programmers who support the open-source-movement philosophy contribute to the open-source community by voluntarily writing and exchanging programming code for software development.

Rights of Users related to Open Source Software:

All open source software grants users the following key rights:

1. The right to full access to the source code. When a computer programmer sees how a piece of software actually works, as specified in the source code, they can fully understand the inner workings and can intelligently modify the software as they deem appropriate.
2. The right for anyone to run the program for any purpose without restriction. There are no restrictions against commercial, military, foreign, or any other use, and discrimination against users for any reason is expressly forbidden.
3. The right to modify the source code. This includes absorbing the software, in whole or in part, into other pieces of software created by other developers.
4. The right to distribute both the original software and the modified software.
5. The right to know about their open source rights: The open source license must be prominently displayed and distributed to users, so that they are aware of their rights (including access to the source code).
6. The obligation to distribute derivatives under copyleft. Any software modified under the open source can be redistributed for sale, but it must be licensed under a copyleft license; that is, modified derivative works must also be made available under an open source license. While it

does not have to be licensed under the GPL itself, the chosen distribution license may not restrict any of the five rights listed above

Evidence of open-source adoption

Libraries:

Libraries are using open-source software to develop information as well as library services. The purpose of open source is to provide a software that is cheaper, reliable and has better quality. The one feature that makes this software so sought after is that it is free. Libraries in particular benefit from this movement because of the resources it provides. They also promote the same ideas of learning and understanding new information through the resources of other people. Open source allows a sense of community. It is an invitation for anyone to provide information about various topics.

Government agencies and infrastructure software — Government Agencies are utilizing open-source infrastructure software, like the Linux operating system and the Apache Web-server into software, to manage information.

Open-source movement in the military — Open-source movement has potential to help in the military. The open-source software allows anyone to make changes that will improve it. This is a form of invitation for people to put their minds together to grow a software in a cost efficient manner. The reason the military is so interested is because it is possible that this software can increase speed and flexibility. Although there are security setbacks to this idea due to the fact that anyone has access to change the software, the advantages can outweigh the disadvantages.

Open source in education — Colleges and organizations use software predominantly online to educate their students. Open-source technology is being adopted by many institutions because it can save these institutions from paying companies to provide them with these administrative software systems.

Open source in healthcare — Created in June 2009 by the nonprofit eHealthNigeria, the open-source software OpenMRS is used to document health care in Nigeria. The use of this software began in Kaduna, Nigeria to serve the purpose of public health. OpenMRS manages features such as alerting health care workers when patients show warning signs for conditions and records births and deaths daily, among other features.

1.13 : Bio-piracy

Biopiracy

‘Biopiracy’ is an emergent term used to name illegal or improper appropriation of traditional knowledge and biological materials the fight against biopiracy,

Vandana Shiva interpreted that, biopiracy is a phenomenon of claiming property rights to biodiversity and its products through intellectual property rights regimes and patents based on indigenous and traditional knowledge

Categories of biopiracy

1 Patent-based biopiracy

The patenting of inventions based on biological resources and/or traditional knowledge that are extracted without adequate authorization and benefit sharing from other countries, indigenous or local communities

2 Non-patent biopiracy

Other intellectual property control based on biological resources and/or traditional knowledge that have been extracted without adequate authorization and benefit-sharing from other (usually developing) countries, indigenous or local communities [

Effects of Biopiracy: □ Examples: Asia: Use to have more than 100,000 varieties of rice in 20th century Now: less than a dozen are planted in 70% of land being cultivated for Rice
India: 30,000 Now: 10 varieties

Protecting Traditional Knowledge from Biopiracy:

The term traditional know-ledge can be categorized into three classes:

Traditional Medicinal Knowledge (TMK),

Traditional Agricultural Knowledge (TAK) and

Traditional Ecological Knowledge (TEK).

Indigenous knowledge is a subset of traditional knowledge category, held and used by communities, peoples and nations. Indigenous people, especially in rural communities, uses Traditional Medicinal Knowledge (TMK) maintain their health systems

Case Examples on Biopiracy:

Recent cases of Bio-piracy

NeemPatent:

US patent office granted patent on a 1 product derived from seeds of the Neem

India opposed the patent by claiming that the properties of the Neem tree had been public knowledge in India for many centuries

The oil from neem has been used traditionally by farmers to prevent fungus. It was neither a novel idea nor was it Inventive.

The Patent was finally revoked by the European Patent Office.

Haldi patent- (Turmeric)

In December 1993, a patent was filed by the University Of Mississippi Medical Center, Mississippi

Applicants received US patent 5,401,504 for the use of turmeric powder as a wound-healing agent.

Indian Government objected to the patent

The turmeric patent failed to satisfy the criteria of novelty in view of the cited turmeric's qualities documented in ancient medical textbooks